

Product Name

ResumeAI Pro

1. Product Overview

Product Type

AI-powered SaaS Web Application

Purpose

Users can create professional ATS-friendly resumes using AI, optimize resumes for specific job descriptions, generate cover letters, and export resumes in PDF format.

The platform should support:

- Students
 - Freshers
 - Professionals
 - Freelancers
 - Remote job seekers
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2. Main Goal

Build a scalable SaaS platform that:

- Generates professional resumes
 - Uses AI for optimization
 - Supports ATS scoring
 - Generates cover letters
 - Supports subscription payments
 - Supports multilingual resumes (English + Bangla)
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3. Core Features

3.1 Authentication System

Features

- Email signup/login
- Google OAuth login
- Forgot password
- Email verification
- JWT authentication
- Session management

User Roles

User

- Create resumes
- Export resumes
- Manage profile

Admin

- Manage users
 - Manage templates
 - View analytics
 - Manage subscriptions
-

3.2 User Dashboard

Dashboard Features

- Total resumes created
- ATS scores
- Subscription status
- Recent resumes

- Cover letters
 - AI credits usage
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3.3 Resume Builder

Resume Sections

Personal Information

- Full name
 - Email
 - Phone
 - Address
 - LinkedIn URL
 - Portfolio URL
 - GitHub URL
 - Profile image
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Professional Summary

AI-generated summary based on:

- Skills
 - Experience
 - Job role
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Work Experience

Fields:

- Company name
- Position
- Start date
- End date
- Current job checkbox
- Responsibilities
- Achievements

AI Enhancement:

- Rewrite bullet points professionally
 - Generate achievement-focused statements
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Education

- Institute name
 - Degree
 - Subject
 - GPA/CGPA
 - Start year
 - End year
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Skills

- Technical skills
- Soft skills
- Languages

AI Suggestion:

- Recommend missing skills based on job role
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Projects

- Project title
 - Description
 - Technologies
 - GitHub link
 - Live link
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Certifications

- Certificate name
- Issuer
- Date

Awards

Optional section

3.4 AI Resume Optimization

User Flow

1. User pastes job description
 2. AI analyzes job description
 3. AI compares resume with JD
 4. AI suggests improvements
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AI Features

- Keyword matching
 - ATS optimization
 - Missing skill suggestions
 - Resume improvement suggestions
 - Bullet point enhancement
 - Grammar fixing
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3.5 ATS Score System

ATS Analysis Parameters

- Keyword match percentage
- Formatting quality
- Readability
- Experience relevance
- Skill match

- Resume length
 - Section completeness
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ATS Score Output

- Overall score (0-100)
 - Detailed breakdown
 - Suggestions list
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3.6 Cover Letter Generator

Features

- AI-generated cover letters
 - Job-specific cover letters
 - Tone selection:
 - Professional
 - Friendly
 - Formal
 - Creative
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3.7 Resume Templates

Template Features

- Multiple professional templates
 - ATS-friendly templates
 - Minimal templates
 - Creative templates
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Customization

- Font selection
 - Font size
 - Theme color
 - Section ordering
 - Spacing adjustments
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3.8 PDF Export

Export Features

- High-quality PDF
 - Multi-page support
 - Print optimization
 - Watermark removal for premium users
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3.9 Multilingual Support

Supported Languages

- English
 - Bangla
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3.10 Subscription & Payment

Plans

Free Plan

- Limited resumes
- Limited AI usage
- Watermarked PDF

Premium Plan

- Unlimited resumes
 - Unlimited exports
 - ATS optimization
 - Cover letter generator
 - Premium templates
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Payment Gateways

- Stripe
 - SSLCommerz
 - PayPal
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3.11 AI Chat Assistant

Features

- Resume writing assistant
 - Career suggestions
 - Skill recommendations
 - Job title recommendations
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3.12 Admin Panel

Features

- User management
- Subscription management
- Revenue analytics
- AI usage analytics

- Template management
 - Content moderation
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4. Technical Requirements

4.1 Frontend

Technology

- Next.js 15
 - TypeScript
 - TailwindCSS
 - ShadCN UI
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Requirements

- Fully responsive
 - SEO optimized
 - Fast loading
 - Dark/light mode
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4.2 Backend

Technology

- Node.js
 - NestJS অথবা Express.js
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Requirements

- REST API
 - JWT authentication
 - Rate limiting
 - API validation
 - Logging
 - Error handling
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4.3 Database

Database

PostgreSQL

ORM

Prisma ORM

4.4 AI Integration

Provider

- OpenAI GPT-4o
 - Claude API fallback
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AI Tasks

- Resume summary generation
- Bullet enhancement
- ATS analysis
- Cover letter generation
- Skill recommendation

4.5 File Storage

Storage Provider

- AWS S3
- or
- Cloudinary

4.6 Deployment

Frontend

- Vercel

Backend

- Railway / Render / AWS

Database

- Supabase PostgreSQL
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5. Database Schema

Users Table

id
name
email
password

role
subscription_plan
created_at
updated_at

Resumes Table

id
user_id
title
template_id
resume_data_json
ats_score
created_at
updated_at

Cover Letters Table

id
user_id
resume_id
content
created_at

Templates Table

id
name
thumbnail
html_structure
is_premium
created_at

Subscriptions Table

id

user_id
payment_provider
plan
status
start_date
end_date

AI Usage Table

id
user_id
feature_used
tokens_used
created_at

6. API Requirements

Authentication APIs

POST /api/auth/register
POST /api/auth/login
POST /api/auth/google
POST /api/auth/forgot-password
POST /api/auth/reset-password

Resume APIs

GET /api/resumes
POST /api/resumes
PUT /api/resumes/:id
DELETE /api/resumes/:id

AI APIs

POST /api/ai/generate-summary
POST /api/ai/optimize-resume
POST /api/ai/generate-cover-letter
POST /api/ai/ats-check

Payment APIs

POST /api/payments/create-checkout
POST /api/payments/webhook
GET /api/subscription/status

7. UI/UX Requirements

Design Style

- Clean
 - Modern
 - Minimal
 - Professional
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Colors

- White
 - Black
 - Blue accent
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UX Requirements

- Drag & drop sections
- Real-time preview
- Auto-save

- Smooth animations
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8. AI Prompt Engineering Requirements

Resume Summary Prompt

Generate a professional resume summary for a [JOB_ROLE] with [YEARS] years of experience using these skills: [SKILLS].

ATS Optimization Prompt

Analyze this resume against the following job description. Suggest improvements, missing keywords, and ATS optimization recommendations.

Cover Letter Prompt

Generate a professional cover letter based on this resume and job description.

9. Security Requirements

Must Implement

- HTTPS
 - JWT expiry
 - Password hashing (bcrypt)
 - Rate limiting
 - Input sanitization
 - XSS protection
 - CSRF protection
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10. Performance Requirements

Requirements

- Page load < 2 seconds
 - API response < 500ms
 - Lighthouse score > 90
 - SEO optimized
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11. SEO Requirements

Pages

- Landing page
 - Blog
 - Resume examples
 - ATS guide
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SEO Features

- Meta tags
 - Sitemap
 - Structured data
 - OpenGraph
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12. Analytics

Integrations

- Google Analytics
- PostHog

Track

- Conversion rate
 - Resume exports
 - User retention
 - AI usage
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13. SaaS Business Logic

Free User Limits

- 2 resumes
 - 5 AI requests/day
 - Watermarked export
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Premium Features

- Unlimited resumes
 - Unlimited AI
 - Premium templates
 - ATS optimization
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14. Future Features

V2 Features

- LinkedIn import
- AI interview preparation
- Job application tracker

- Resume website generator
 - Team accounts
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15. Recommended Folder Structure

```
src/  
├── app/  
├── components/  
├── modules/  
├── services/  
├── lib/  
├── hooks/  
├── prisma/  
├── types/  
├── utils/  
└── styles/
```

16. Development Instructions for AI Coding Agents

Important Rules

Code Quality

- Use TypeScript everywhere
 - Reusable components only
 - Proper folder architecture
 - Clean code principles
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Backend Rules

- Use service-based architecture
- Use DTO validation
- Use Prisma transactions

Frontend Rules

- Responsive design mandatory
 - Use server components where possible
 - Use React Query for API state
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AI Integration Rules

- Token optimization
 - Retry mechanism
 - Streaming responses if possible
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17. Exact Tech Stack

Layer	Technology
Frontend	Next.js
Backend	NestJS
Database	PostgreSQL
ORM	Prisma
Auth	NextAuth
AI	OpenAI
Storage	AWS S3
Payments	Stripe
Deployment	Vercel

18. Landing Page Sections

- Hero section
 - Features
 - Templates showcase
 - ATS optimization demo
 - Pricing
 - Testimonials
 - FAQ
 - Footer
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19. Monetization Strategy

Revenue Sources

- Monthly subscription
 - Yearly subscription
 - Resume template marketplace
 - Career coaching upsell
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20. Final Product Goals

The final product should:

- Be production-ready
 - Be scalable
 - Be SaaS optimized
 - Be mobile responsive
 - Support global users
 - Generate recurring revenue
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END OF REQUIREMENTS DOCUMENT